

Profit & loss

COMPLETE LEARNING GUIDE

What You'll Learn

Profit and loss questions test whether a candidate can correctly identify which value — cost price, marked price, or selling price — a given percentage is calculated against, since mixing these up is the single most common source of wrong answers on this topic. A markup is always applied to cost price, a discount is always applied to marked price, and profit/loss percentage is always calculated against cost price — three different bases that must never be confused. This sub-module works through the standard CP-SP-profit% relationship, the marked-price-and-discount combination (including the classic 'successive markup then discount' trap), false-weight dishonest-dealer problems, multi-stage transactions, partnership profit-sharing, and CP problems that include overhead expenses — each with a complete numerical solution so the underlying base-value logic, not just the final formula, becomes second nature.

1. Core CP-SP-Profit% Relationship

Profit = SP - CP, and Profit% is always calculated on Cost Price: $\text{Profit\%} = (\text{Profit}/\text{CP}) \times 100$. Loss% follows the same logic with Loss = CP - SP. Memorizing 'percentage is on CP, always' prevents the most common error on this topic.

- CP = ₹800, SP = ₹920. Profit = ₹120. $\text{Profit\%} = 120/800 \times 100 = 15\%$.

2. Marked Price, Discount, and the Successive-Change Trap

Marked price (MP) is set above CP by a markup percentage; the actual selling price after a discount is applied to MP, not to CP. A markup followed by a discount is a successive percentage change (see the Percentages sub-module) and must never be treated as canceling out.

- A shopkeeper marks goods up 40% then gives a 25% discount. Let CP = 100. MP = 140. SP = $140 \times 0.75 = 105$. Net profit% = 5% — not a wash, despite the discount percentage exceeding the markup by nothing obvious at a glance.

3. False Weight (Dishonest Dealer) Problems

When a dealer sells at cost price but uses a weight less than what's claimed, the dealer's real profit% = $(\text{True weight} - \text{Used weight})/(\text{Used weight}) \times 100$ — the profit comes purely from the weight shortfall, not from any price markup.

- A dealer claims to sell at cost price but uses a 900 g weight for a claimed 1 kg. Real profit% = $(1000 - 900)/900 \times 100 = 11.11\%$.

4. Successive Transactions (Buying and Reselling)

When an item changes hands more than once, each sale's profit/loss percentage applies to that transaction's own CP (the previous SP), not to the original CP — chain the transactions sequentially rather than adding percentages.

- A buys an item for £1000 and sells to B at 20% profit: SP to B = £1200. B then sells to C at a 10% loss: SP to C = $1200 \times 0.9 = £1080$. Overall, £1000 became £1080 — a net 8% profit over the two transactions (not 20% - 10% = 10%).

5. Partnership Profit Sharing

When partners invest different amounts for different durations, profit is shared in the ratio of (investment $\times$ time) for each partner, not simply the investment ratio alone.

- A invests £5000 for 12 months, B invests £6000 for 8 months. Ratio = $(5000 \times 12) : (6000 \times 8) = 60000 : 48000 = 5 : 4$. If total profit is £9000, A gets £5000 and B gets £4000.

6. Reverse Problems: Finding MP or CP From SP

Many exam questions give the final SP and a discount or profit percentage and ask you to work backward to MP or CP — this requires dividing by $(1 \pm \text{percentage})$, not multiplying, since the percentage was originally applied forward to the unknown value.

- An item sells for £450 after a 10% discount. $MP \times 0.9 = 450$, so $MP = 450/0.9 = £500$.

7. Cost Price Including Overhead Expenses

Real CP for profit% calculations must include all expenses incurred before sale (repairs, transport, packaging) — using only the purchase price and ignoring overheads understates the true cost base and overstates the profit percentage.

- An article is bought for £1200, with £300 spent on repairs, then sold for £1800. True CP = $1200 + 300 = £1500$. Profit = $1800 - 1500 = £300$. Profit% = $300/1500 \times 100 = 20\%$ (not $300/1200 = 25\%$, which ignores the overhead).

Common Pitfalls

- Calculating discount on CP instead of MP, or markup on SP instead of CP — always: markup is on CP, discount is on MP.
- Assuming a markup% and a larger discount% of the same numeric value cancel to a net loss — always compute using $\text{net}\% = a + b + ab/100$.
- In false-weight problems, calculating profit% on the claimed weight instead of the actually-used (smaller) weight.

- Adding profit and loss percentages across successive transactions instead of chaining each one onto the previous SP.
- Using investment amount alone for partnership profit-sharing, ignoring the time each partner's capital was invested.
- Forgetting to include overhead/incidental expenses in CP before calculating profit percentage.
- When working backward from SP to find MP or CP, subtracting the percentage instead of dividing by $(1 \pm \text{percentage}/100)$.

Key Takeaways

- Profit% and Loss% are always calculated on Cost Price — never on SP or MP.
- Markup applies to CP to get MP; discount applies to MP to get SP — two different base values, never interchange them.
- False-weight profit% = $(\text{claimed weight} - \text{used weight}) / (\text{used weight}) \times 100$.
- Chain successive transactions sequentially (each SP becomes the next CP) rather than adding percentages.
- Partnership profit ratio = (investment $\times$ time) for each partner, not investment alone.
- To reverse-solve for MP/CP from a known SP, divide by $(1 \pm \text{rate}/100)$, don't just subtract the percentage.

Learn More

IndiaBix — Profit and Loss Aptitude Questions

<https://www.indiabix.com/aptitude/profit-and-loss/>

Investopedia — Gross Profit Margin

<https://www.investopedia.com/terms/g/grossprofitmargin.asp>